

Colin Hanley

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EDUCATION

University of Alabama

Bachelor of Science in Mathematics and Data Science

Tuscaloosa, AL

Aug. 2023 – May 2027*

- GPA: 3.94/4.00
- Relevant Coursework: Machine Learning, Mathematical Foundations of Data Science, Data Structures and Algorithms, Theory of Probability, Quantum Computing

EXPERIENCE

Undergraduate Research Assistant

University of Alabama

Jan. 2025 – Present

Tuscaloosa, AL

- Designed and fine-tuned LLM-based agents to forecast groundwater depth patterns across the U.S.
- Implemented training pipelines using Python, Pytorch, and HuggingFace Transformers
- Preprocessed and validated large-scaled geospatial datasets to ensure model integrity

PSIA-Certified Snowsports Instructor

Lost Trail Powder Mountain

Dec. 2021 – Jan. 2025

Sula, MT

- Designed and implemented lesson plans for beginner and intermediate snowsports students
- Assisted in the training of new snowsports instructors entering the program
- Spearheaded improvements to the lesson-assignment process, enhancing scheduling efficiency and instructor coordination

PROJECTS

HL7 Parser and Deidentifying Tool(UA Innovate Hackathon 2nd place) | *Python, Streamlit*

- Engineered a Python-based tool for parsing and anonymizing HL7-formatted medical records to ensure HIPAA compliance
- Created a front-end application using Streamlit to display this data and allow for searching and filtering

Agent-Based COVID-19 Epidemic Simulator | *Python, Mesa*

- Developed an agent-based simulation in Python to analyze COVID-19 spread, enabling scenario testing for public health interventions like vaccine strategies, masking policy, and lockdown thresholds
- Engineered features including household/workplace contact structures, age-based risk profiles, detailed immunity profiles and dynamic lockdown triggers
- Utilized Mesa for model architecture, scheduling, spatial grid, and interaction web visualization of simulation outputs

Poacher Detection in Infrared Images | *Python, Pytorch, YOLOv5*

- Fine-tuned a custom YOLOv5 model for real-time detection of human presence in LWIR UAV imagery
- Achieved 76% recall on the targeted human class after writing a script to pseudo-label synthetic data
- Wrote and presented an academic paper discussing results

TECHNICAL SKILLS

Languages: Python, R, MATLAB, C++, SQL

Libraries: Pytorch, Scikit-learn, NumPy, Pandas